

Code

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1a.

```
#load in the many libraries
library(ggplot2)
library(dplyr)
library(matrixStats)
library(gt)
library(gtExtras)
# Make results reproducible
set.seed(5771)

#NOTE: I just reused this for every sample size and changed the title in the ggplot

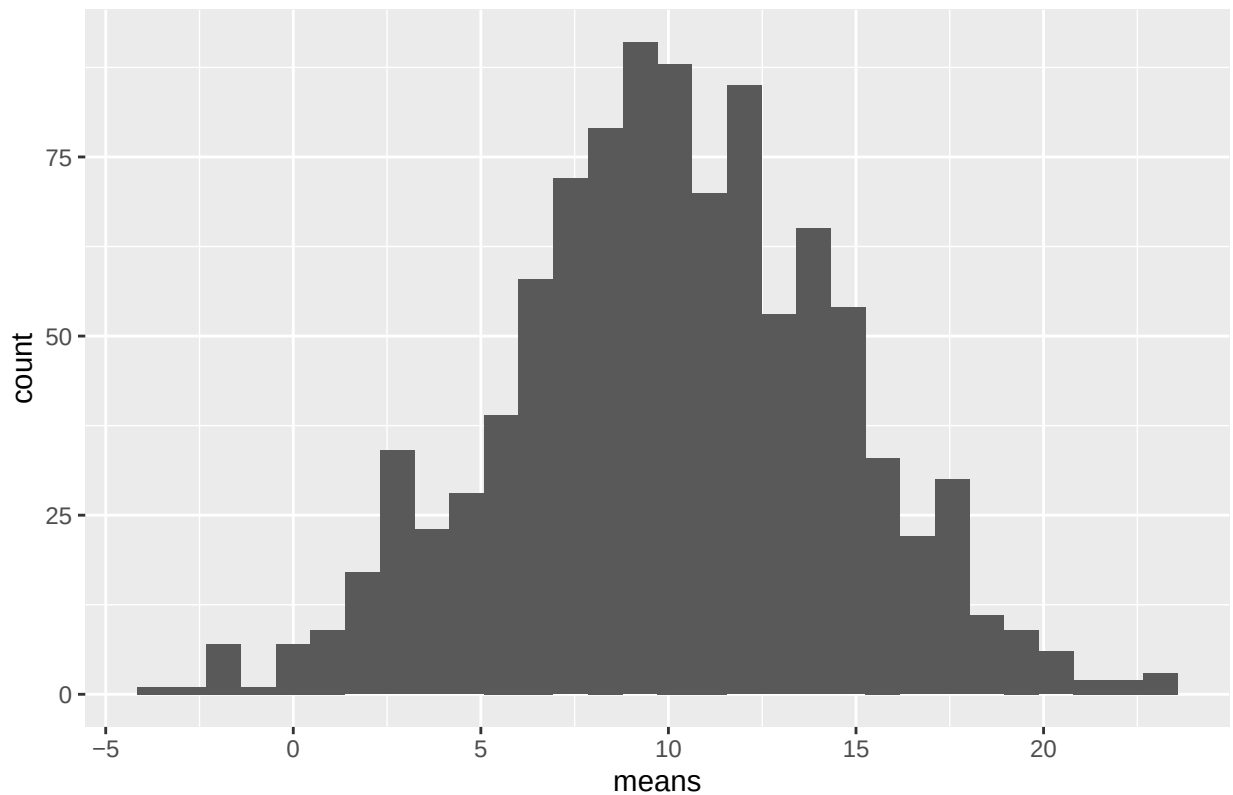
# Define sample size
n <- 5 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#graphing
ggplot(data = normal_means) + geom_histogram(aes(x = means)) +
  labs(title = "Histogram of Means for 1000 Samples: Size 5")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Histogram of Means for 1000 Samples: Size 5



1b.

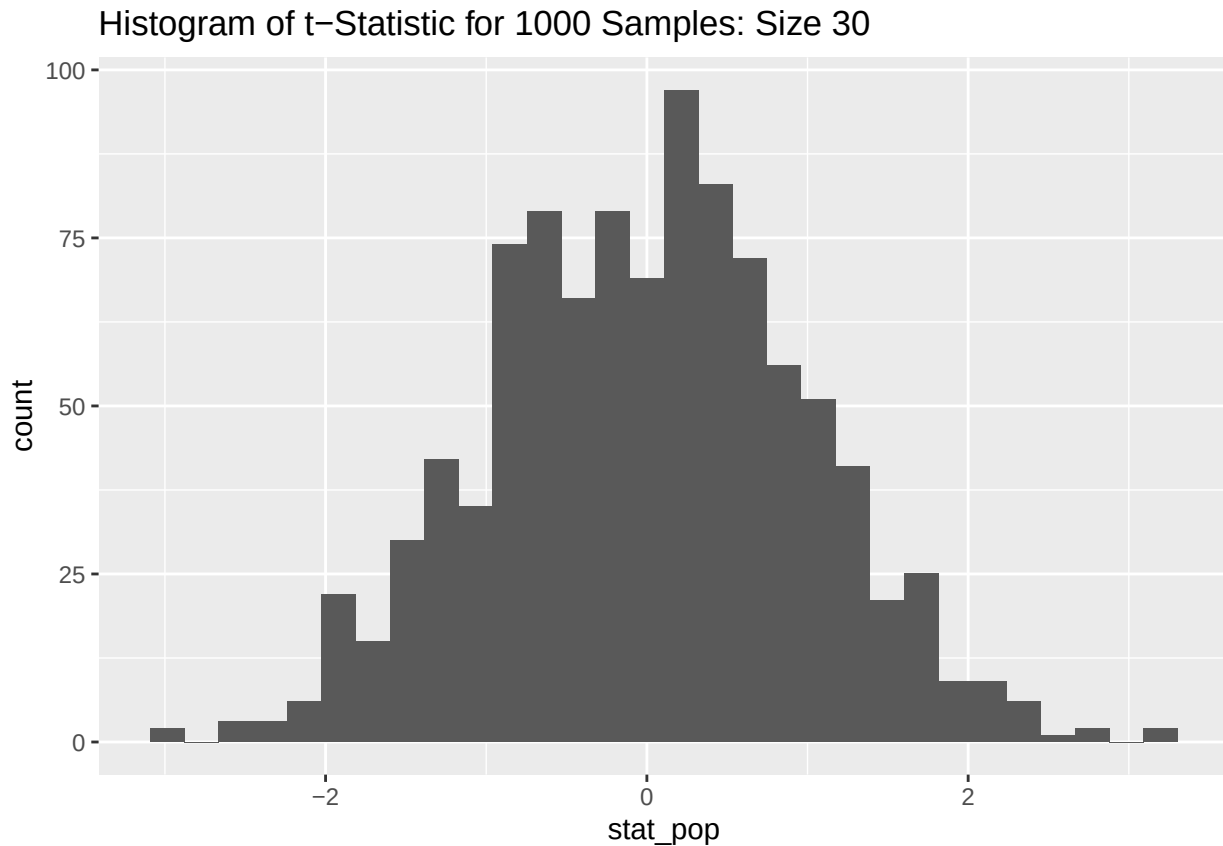
```
n <- 5 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#calculating the t statistic with the pop sd
normal_means <- normal_means %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
```

```
#graph
ggplot(data = normal_means) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Histogram of t-Statistic for 1000 Samples: Size 30")
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



1c.

```
n <- 500 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
```

```

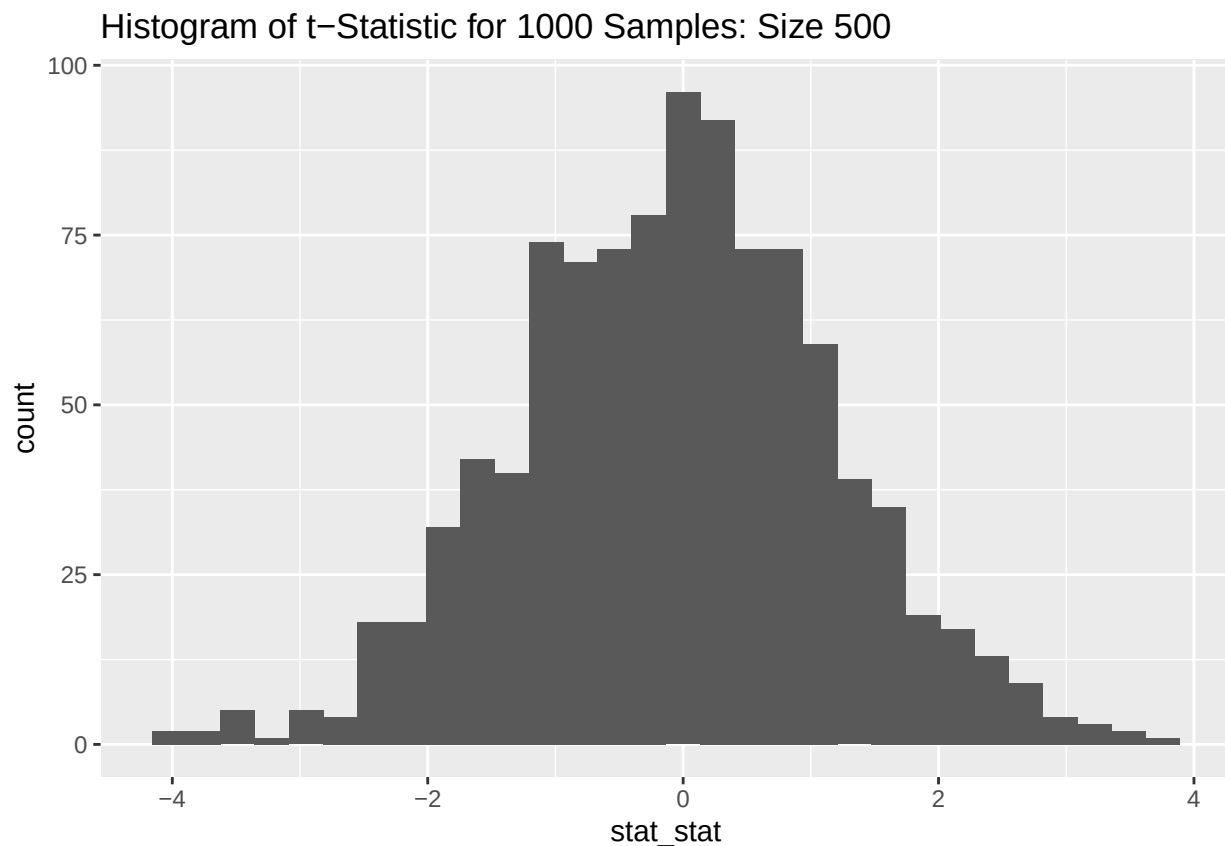
#rename the column
colnames(normal_means)[2] <- "sd"

#calculating the t statistic with the pop sd
normal_means <- normal_means %>%
  mutate(stat_stat = ((means - sd)/(10/sqrt(n))))

#graph
ggplot(data = normal_means) + geom_histogram(aes(x = stat_stat)) +
  labs(title = "Histogram of t-Statistic for 1000 Samples: Size 500")

```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



d.

```

n <- 500 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df

```

```

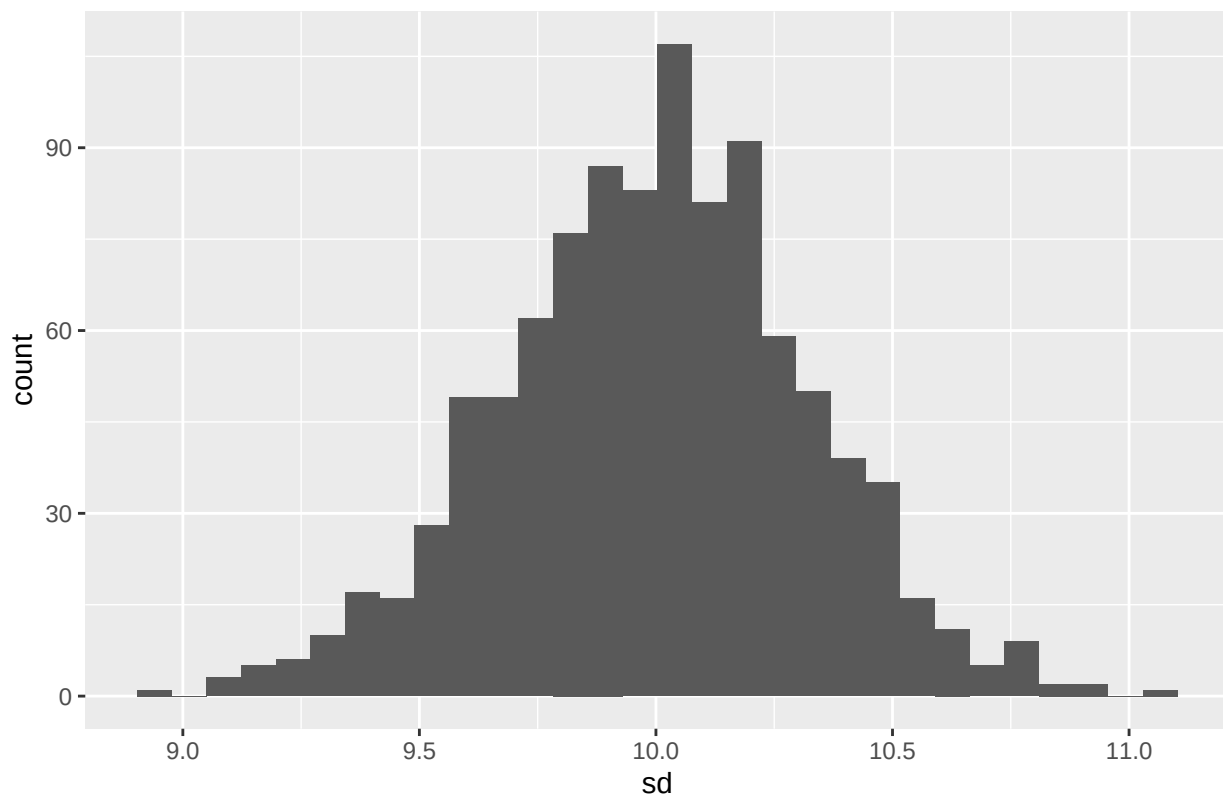
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#graph
ggplot(data = normal_means) + geom_histogram(aes(x = sd)) +
  labs(title = "Histogram of SD for 1000 Samples: Size 500")

```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.

Histogram of SD for 1000 Samples: Size 500



e.

```

#NOTE:ran this iteratively for all the sample sizes and then used both the
#pop sd and the sample sd

n <- 500 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10

```

```

sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#calculating the t statistic with the pop sd
normal_means <- normal_means %>%
  mutate(stat_pop = ((means - sd)/(10/sqrt(n))))
#check if our test statistic is farther that 1.96 away!
passes_popsd <- normal_means %>%
  #take abs bc it has to be 1.96 away in either direction
  mutate(teststatistic = abs(stat_pop))
#count how many are greater than 1.96
nrow(passes_popsd[passes_popsd$teststatistic>1.648, ])

```

```
## [1] 174
```

le.

```

#pretty table time LETS GOOOO
names <- NA
n_5 <- NA
n_10 <- NA
n_30 <- NA
n_100 <- NA
n_500 <- NA

table <- data.frame(names, n_5, n_10, n_30, n_100, n_500)
table[1, ] <- list("Population SD", "56", "40", "42", "43", "52")
table[nrow(table) + 1, ] <- list("Sample SD", "114", "113", "115", "112", "113")
table[nrow(table) + 1, ] <- list("t-distribution and Sample SD", "93", "153", "184", "176", "174")

gt_table <- gt(table)
gt_table <- gt_table %>%
  cols_label(
    names = "Sample Size:",
    n_5 = "5",
    n_10 = "10",
    n_30 = "30",
    n_100 = "100",
    n_500 = "500"
  )

```

```

) %>%
gt_add_divider(columns = "names")

gt_table

```

Sample Size:	5	10	30	100	500
Populatin SD	56	40	42	43	52
Sample SD	114	113	115	112	113
t-distribution and Sample SD	93	153	184	176	174

```

names <- NA
n_5 <- NA
n_10 <- NA
n_30 <- NA
n_100 <- NA
n_500 <- NA

table <- data.frame(names, n_5, n_10, n_30, n_100, n_500)
table[1, ] <- list("df", "4", "9", "29", "99", "499")
table[nrow(table) + 1, ] <- list("Critical Value", "2.132", "1.833", "1.699", "1.660", "1.648")

gt_table <- gt(table)
gt_table <- gt_table %>%
  cols_label(
    names = "Sample Size:",
    n_5 = "5",
    n_10 = "10",
    n_30 = "30",
    n_100 = "100",
    n_500 = "500"
  ) %>%
  gt_add_divider(columns = "names")

gt_table

```

Sample Size:	5	10	30	100	500
df	4	9	29	99	499
Critical Value	2.132	1.833	1.699	1.660	1.648

Part 2

```

# Define sample size
n <- 10 # you'll only try 10, 30, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 2: gamma
gamma_data <- replicate(1000, rgamma(n, shape = mu^2/sigma^2, scale = sigma^2/mu))
#taking the means and turning into a df

```

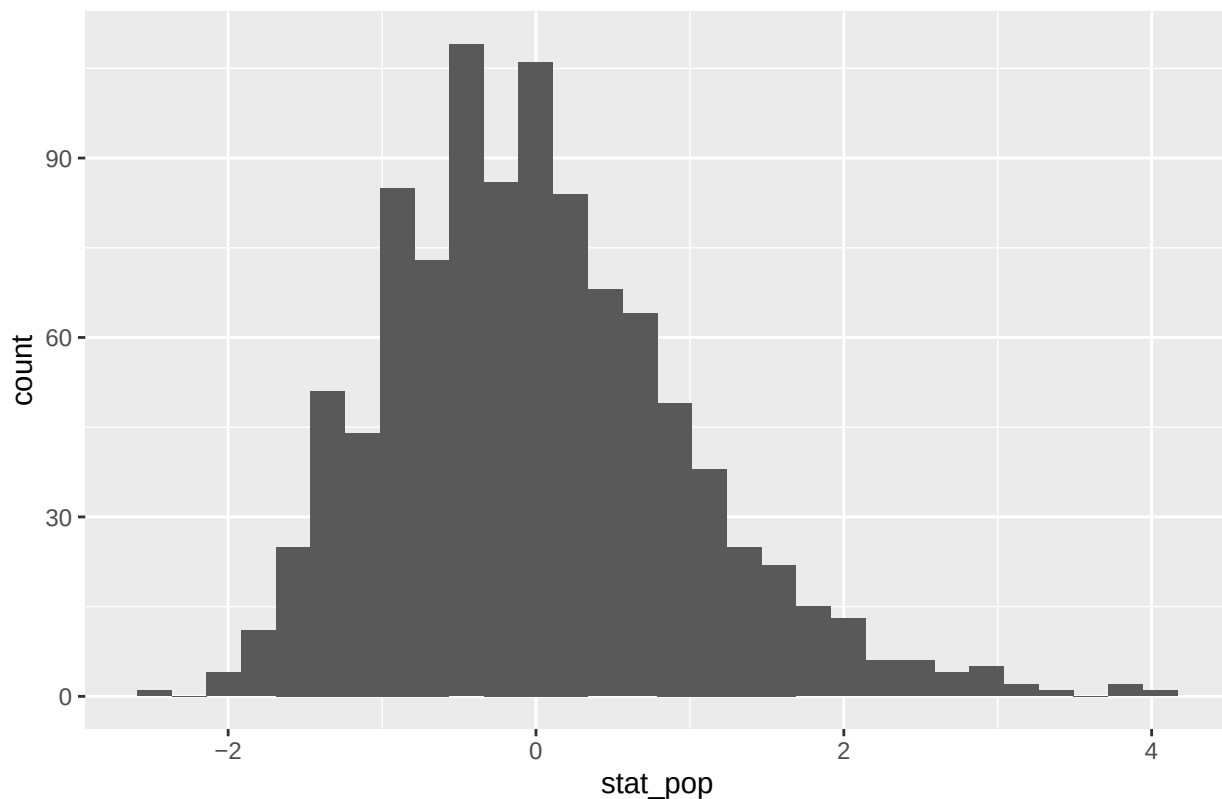
```

gammadata <- data.frame(colMeans(gamma_data))
#changing the name of the column
colnames(gammadata)[1] <- "means"
#finding the sd of
sd <- colSds(gamma_data)
#adding the sd to our means df
gammadata <- gammadata %>%
  cbind(colSds(gamma_data))
#rename the column
colnames(gammadata)[2] <- "sd"
gammadata <- gammadata %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
#graph
ggplot(data = gammadata) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Gamma Distribution Test Statistic: Sample Size 500")

```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Gamma Distribution Test Statistic: Sample Size 500



```

passes_popsd <- gammadata %>%
  #take abs bc it has to be 1.96 away in either direction
  mutate(teststatistic = abs(stat_pop))
#count how many are greater than 1.96
nrow(passes_popsd[passes_popsd$teststatistic>1.833, ])

```

```
## [1] 51
```



```

# Population 3: using the uniform, but we need them to have the same mu and sigma

n <- 10 # you'll only try 10, 30, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 2: gamma
#so i calculated the needed min and max to get 10 and 10
data3 <- replicate(1000, runif(n, min= -7.32, max =27.32))
#taking the means and turning into a df
uniformdata <- data.frame(colMeans(data3))
#changing the name of the column
colnames(uniformdata)[1] <- "means"
#finding the sd of
sd <- colSds(data3)
#adding the sd to our means df
uniformdata <- uniformdata %>%
  cbind(colSds(data3))
#rename the column
colnames(uniformdata)[2] <- "sd"
uniformdata <- uniformdata %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
#graph
ggplot(data = uniformdata) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Uniform Distribution Test Statistic: Sample Size 10")

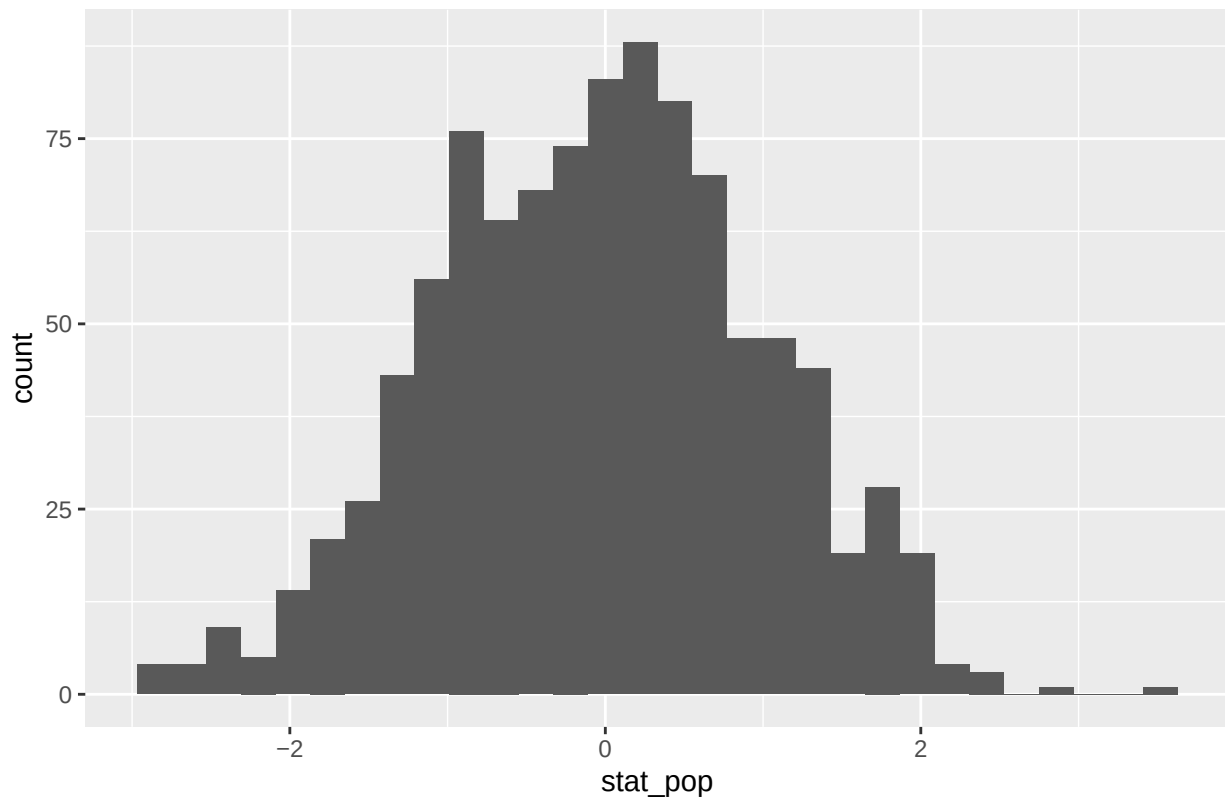
```

```

## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.

```

Uniform Distribution Test Statistic: Sample Size 10



```
passes_popsd <- uniformdata %>%
  #take abs bc it has to be 1.96 away in either direction
  mutate(teststatistic = abs(stat_pop))
#count how many are greater than 1.96
nrow(passes_popsd[passes_popsd$teststatistic>1.648, ])
```

```
## [1] 113
```

```
#pretty table time LETS GOOOO
names <- NA
n_10 <- NA
n_30 <- NA
n_500 <- NA

table <- data.frame(names, n_10, n_30, n_500)
table[1, ] <- list("Normal Distribution", "153", "184", "174")
table[nrow(table) + 1, ] <- list("Gamma Distribution", "66", "74", "118")
table[nrow(table) + 1, ] <- list("Uniform Distribution", "70", "77", "104")

gt_table <- gt(table)
gt_table <- gt_table %>%
  cols_label(
    names = "Sample Size:",
    n_10 = "10",
    n_30 = "30",
```

```

n_500 = "500"
) %>%
gt_add_divider(columns = "names")
gt_table

```

Sample Size:	10	30	500
Normal Distribution	153	184	174
Gamma Distribution	66	74	118
Uniform Distribution	70	77	104